

1. Linear Motion of Bodies

1.1 Displacement, Velocity and Acceleration.

Displacement: Can be defined as a distance from one point to another in a specified direction. Displacement is a vector quantity and is measured in meters (m)

Velocity: Can be defined as the rate of change of displacement with respect to time. Velocity is a vector quantity and is mathematically defined as

$$\text{Velocity} = \frac{\text{displacement}}{\text{time}}$$

The S.I unit of velocity is meters per second ie m/s

Acceleration: Can be defined as the rate of change of velocity with respect to time. Acceleration is a vector quantity and is mathematically defined as:

$$\text{acceleration} = \frac{\text{velocity}}{\text{time}}$$

S.I unit of acceleration is m/s^2

Relationship between Displacement, velocity and Acceleration:

$$\text{Velocity (V)} = \frac{\text{displacement (S)}}{\text{time (t)}} \quad \text{--- (i)}$$

$$\text{Acceleration (a)} = \frac{\text{velocity (V)}}{\text{time (t)}} \quad \text{--- (ii)}$$

$$\text{From (i)} \quad t = \frac{S}{V} \quad \text{--- (iii)}$$

Substitute (iii) in (ii)

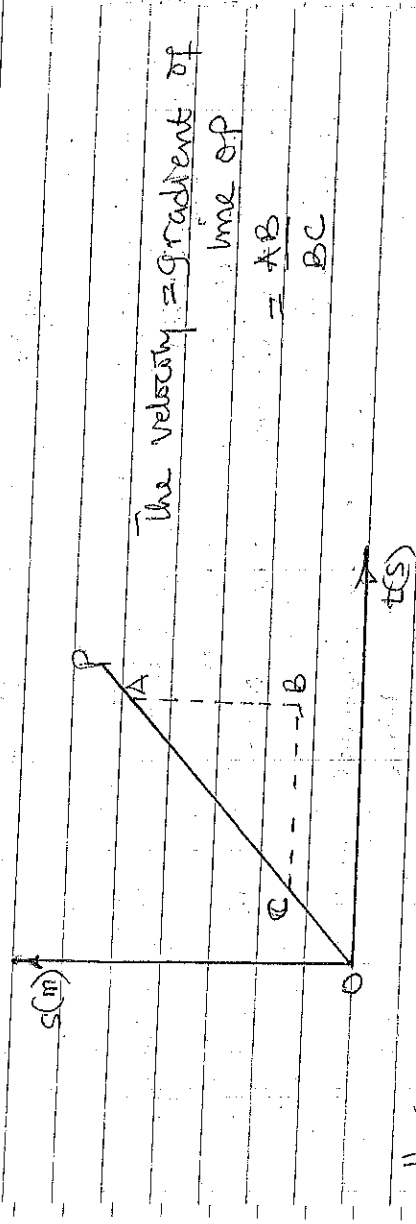
$$a = \frac{V}{S/V} \Rightarrow a = \frac{V^2}{S}$$

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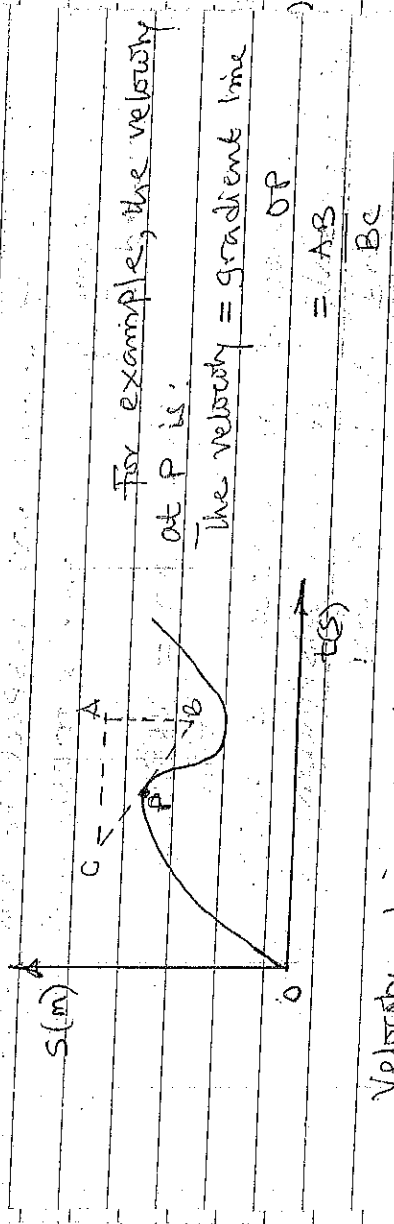
Displacement - time graphs.

In mechanics, the displacement - time graph provide useful information on motion of an object. For instance, If the displacement - time graph is a straight line, it means the velocity is uniform.

The velocity is the slope or gradient.

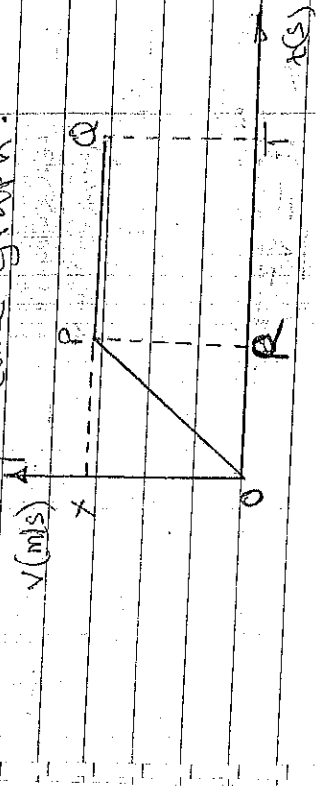


However, if the motion is not a straight line, the velocity at any point is the gradient of the tangent of that particular point.



Velocity - time graph.

Velocity - time graph is the magnitude of acceleration at any point of the motion can be found using the velocity - time graph.



(3)

The acceleration (a) of the motion OP is determined by finding the gradient of line OP, thus,
Acceleration (a) of OP = gradient of line OP

$$= \frac{PR}{OR}$$

Similarly, the acceleration of the motion PQ is also determined by finding the gradient of the line PQ, thus,

$$\text{Acceleration (a) of PQ} = \text{gradient of line PQ} \\ = 0$$

Note that, the motion from P to Q is at constant velocity when velocity is constant, the acceleration must be zero.
Uniform Acceleration: can be defined as the motion of an object whose velocity increases by equal amounts in equal time intervals, however small the time interval may be.

Worked Example 1

Q1 A car travelling at 72 km/h accelerates uniformly at 3 m/s^2 . find its velocity in 10 seconds.

Solution:

Initial velocity, $u = 72 \text{ km/h}$

$$= \frac{72 \times 1000}{60 \times 60} = \underline{\underline{20 \text{ m/s}}}$$

Acceleration (a) = 3 m/s^2

Time (t) = 10 seconds

Final vel. (v) = ?

Acceleration $a = v - u$

$$v = u + at = 20 + 3(10) \\ = \underline{\underline{50 \text{ m/s}}}$$

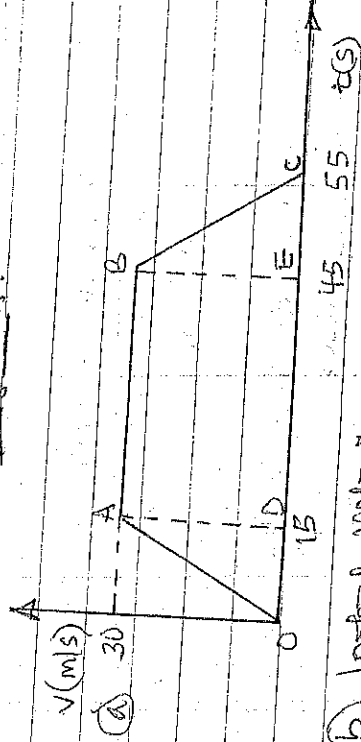
Q2 A train starting from rest accelerates uniformly at 3 m/s^2 for 15 seconds. It then travels for 30 seconds at constant velocity and is brought to rest with uniform deceleration after 10 seconds.

(a) Draw a velocity-time graph of the motion.

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- Find the velocity reached.
- Find the distance travelled during the acceleration.
- Find the distance travelled during the constant velocity.
- Find the total distance travelled.
- Find the deceleration.

Solution:



(b) Initial velocity, $u = 0 \text{ m/s}$

Acceleration, $a = 2 \text{ m/s}^2$

time, $t = 15 \text{ sec}$

Final velocity, $v = ?$

The acceleration, a , is given by

$$a = \frac{v - u}{t} \Rightarrow v = u + at$$

$$= 0 + 2(15)$$

$$= 30 \text{ m/s}$$

(c) Distance travelled during acceleration = Area of $\triangle AOD$

$$= \frac{1}{2} \times 15 \times 30$$

(d) Distance travelled during constant velocity =

Area of rectangle ABDE

$$= 30 \times 30$$

$$= 900 \text{ m}$$

(e) Distance travelled during deceleration =

Area of $\triangle BCE$

$$= \frac{1}{2} \times 10 \times 30 = 150 \text{ m}$$

(f) Total distance travelled = Area of trapezium OABC

$$= \frac{1}{2} (55 + 30) \times 30$$

(5)

$$= 1,275 \text{ m}$$

The deceleration is the slope on line BC = $30 - 0$

$$45 - 55$$

$$= -3$$

$\therefore$ the deceleration is 3 m/s^2

Motion Under Gravity:

A body moving in space under the influence of gravitational force is said to move with uniform acceleration. This uniform acceleration under this condition is called the gravitational acceleration denoted by " g ".

When an object is projected upward the gravitational acceleration g is considered to be negative because the object is moving against gravity, similarly, when an object is dropped from a certain height, the gravitational acceleration g is positive because the object is moving towards gravity.

In ~~some~~ Problems of motion under gravity, the linear acceleration a is replaced by gravitational acceleration (g) in motion under gravity.

Linear Motion	Motion Under Gravity	
(i) $V = u + at$	Downward motion	Upward motion
(ii) $S = ut + \frac{1}{2}at^2$	$V = u + gt$ $S = ut + \frac{1}{2}gt^2$	$V = u + (-g)t$ $S = ut + \frac{1}{2}(-g)t^2$
(iii) $V^2 = u^2 + 2as$	$V^2 = u^2 + 2gs$	$V^2 = u^2 + 2(-g)s$

Worked Example

Q1. A ball is thrown upward with a velocity of 12 m/s . How high does it reach and how long does it take to get there (Take $g = 10 \text{ m/s}^2$)

Solution:

Initial Velocity, $u = 12 \text{ m/s}$

Acceleration $g = 10 \text{ m/s}^2$

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At the maximum height S , the final velocity, $v=0$, from the equation of motion

$$v^2 = u^2 + 2as, \text{ to determine the value of } S$$

$$0^2 = u^2 + 2(-g)S$$

$$0^2 = 12^2 + 2 \times (-10) \times S$$

$$S = 7.2 \text{ m}$$

g is -ve because the ball is thrown upward i.e. the ball is moving against gravity.

The time taken (t) for the ball to reach the maximum height can be determined from the equation

$$v = u + gt$$

$$0 = 12 + (-10) \times t$$

$$t = 1.2 \text{ sec.}$$

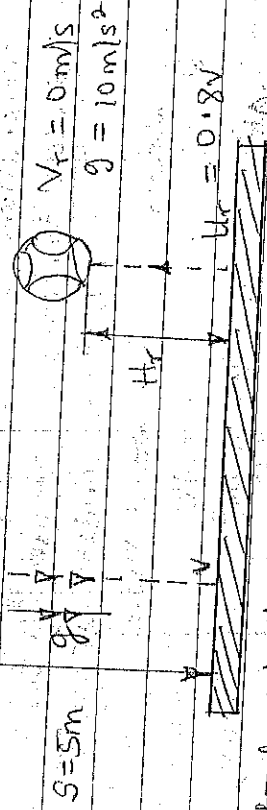
Q2 A ball is dropped from a height of 5m onto a concrete floor and rebounds with a speed 0.8 times the downward speed on arrival.
Find the height from the rebound (Take $g = 10 \text{ m/s}^2$) and sketch the displacement

Solution



$$u = 0 \text{ m/s}$$

$$g = 10 \text{ m/s}^2$$



Let u = Initial velocity at which the ball is dropped
 v = final velocity at which the ball reached the concrete floor

u_r = Initial velocity of the rebound
 S = height at which the ball is dropped
 v_r = final vel. at which maximum rebound height (h_r)

g = gravitational acceleration

To find the velocity, v , at which the ball reached

Contd.

The concrete floor.

Use the equation of motion

$$V^2 = u^2 + 2gs$$
$$= 0^2 + 2 \times 10 \times 5$$

$$V^2 = 100 \text{ m/s}^2 \Rightarrow V = 10 \text{ m/s}$$

From the sketch, the initial velocity of the rebound u_r is equal to 0.8 times the velocity V at which the ball reached the concrete floor,

This implies,

$$u_r = 0.8 \times V$$
$$= 0.8 \times 10$$

$$\therefore u_r = 8 \text{ m/s}$$

To determine the maximum height of rebound H_r , use the equation of motion

$$V^2 = u^2 + 2gs$$

$$\Rightarrow V^2 = u^2 + 2gH_r$$

$$0^2 = 8^2 + 2 \times (-10) \times H_r$$

$$H_r = \underline{\underline{3.2 \text{ m}}}$$

⑦

$$v = u + at$$

$$\text{But } t = 20 \text{ sec}$$

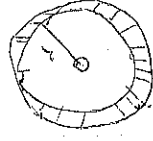
$$v = 5 + 0.4 \times 20$$

$$= \underline{13 \text{ m/s}}$$

The distance (s) travelled in 12 sec can be determined from equation,

$$s = \frac{1}{2}(u+v)t$$

$$= \frac{1}{2}(5+13) \times 20 = \underline{180 \text{ m}}$$



~~Angular velocity~~

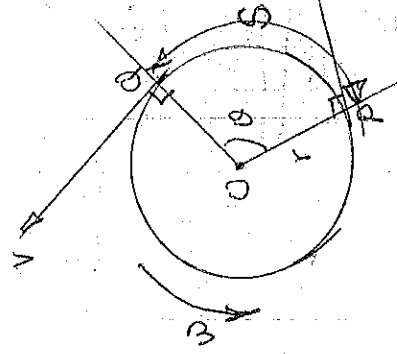
Curvilinear Motion of Bodies:

Relationship b/w Angular and Linear motion

Consider a wheel of radius r , rotating about center O with uniform angular velocity ω rad/s.

$$v = \text{linear velocity} = \text{m/s}$$

$$\omega = \text{angular} \quad \checkmark = \text{rad/s}$$



Let the initial position of the point be at point P and after time t (sec), let it be at Q .

The radius OP would then have turned through an angle $\theta = \omega t$ rad. In this time, the point will have moved through the arc PQ of length s ,

$$\therefore s = r\theta \quad \theta = \frac{s}{r} \quad \text{— angular displacement } (\theta)$$

$$\text{But } \theta = \omega t \quad \text{— Linear displacement } (s)$$

$$\Rightarrow s = r\omega t \quad \text{— (i)}$$

Note!

$$S = \frac{\theta}{180} \times \pi r$$
$$= r \theta \times \frac{\pi}{180}$$

But θ is to be expressed in radian

$$\therefore S = r \theta \quad \left(1 \text{ rad} = \frac{\pi}{180}\right)$$

$$\text{Linear velocity (V)} = \frac{\text{distance moved (S)}}{\text{time taken (t)}}$$

$$\Rightarrow V = \frac{S}{t} \quad \text{(i)}$$

But from equation (i), $S = r \omega t$

Substitute in eqn. (i) above

$$\boxed{V = r \omega t} = r \omega$$

Linear velocity = radius $\times$ Angular Velocity.
in time t (sec):

Angular acceleration (α) = $\frac{\text{change in angular velocity}}{\text{time taken}}$

$$\Rightarrow \alpha = \frac{\omega_1 - \omega_0}{t}$$

$$\text{But } \omega_1 = \frac{V_1}{r} \text{ and } \omega_0 = \frac{V_0}{r}$$

$$\alpha = \frac{\frac{V_1}{r} - \frac{V_0}{r}}{t} = \frac{1}{r} \times \frac{V_1 - V_0}{t} \quad \text{--- (1)}$$

But linear acceleration, $a =$

$$\Rightarrow a = \frac{1}{r} \times a$$

$$\Rightarrow \boxed{a = \alpha r}$$

∴ Linear acceleration = radius $\times$ angular acceleration (9)

Worked Examples

1. A grinding wheel has a diameter of 400 mm. If it rotates at 2000 rev/min, what is the circumferential grinding speed in m/s.

Solution:

Diameter of grinding wheel, $D = 400 \text{ mm}$
 $= \underline{0.4 \text{ m}}$

∴ Radius of the grinding wheel $\Rightarrow r = \frac{D}{2} = \frac{0.4}{2} = \underline{0.2 \text{ m}}$
 Speed of rotation $N = 2000 \text{ rev/min}$

Angular velocity, $\omega = \frac{2\pi N}{60}$

$$= \frac{2 \times \pi \times 2000}{60} = 209.46 \text{ rad/sec}$$

Circumferential speed = Tangential speed $= v$
 $v = r\omega$

$$= 0.2 \times 209.46$$

$$= \underline{41.8 \text{ m/s}}$$

2. A vehicle with wheels $\frac{750 \text{ mm}}{2}$ in diameter moving at 27 km/h is brought to rest with uniform retardation to a distance of 60 m. Find:

- The retardation of the vehicle, in m/s^2
- The initial angular velocity of the wheel, in rad/sec .
- The angular retardation of the wheel, in rad/sec^2

Solution:

Radius of wheel, $r = \frac{0.750}{2}$

$$= \underline{0.375 \text{ m}}$$

Initial linear velocity, $u = \frac{27 \text{ km/h}}{3.6}$

$$= \frac{27 \times 1000}{60 \times 60} = \underline{7.5 \text{ m/s}}$$

Final linear velocity, $v = 0 \text{ m/s}$
Distance travelled, $s = 60 \text{ m}$

(10)

(a) To find the retardation of the vehicle

$$v^2 = u^2 + 2as$$

$$(0)^2 = (7.5)^2 + 2 \times a \times 60$$

$$\therefore a = -\underline{\underline{0.468 \text{ m/s}^2}}$$

b. Initial linear velocity, $u =$ radius of wheel $\times$ initial angular velocity ω_1

$$u = r \times \omega_1$$

$$\omega_1 = \frac{u}{r} = \frac{7.5}{0.375}$$

$$\omega_1 = \underline{\underline{20 \text{ rad/s}}}$$

(c) linear retardation, $a =$ radius of wheel $\times$ angular retardation (α)

$$a = r \times \alpha$$

(3.) A car wheel of 0.6 m diameter is uniformly accelerated in a straight line from 35 km/h to 60 km/h in 8 sec. cal.

(a) The angular acceleration of the wheel.

(b) The number of revolutions made by the wheels during acceleration.

Solution:

Convert the linear velocities from km/h to m/s

$$u_1 = \frac{35 \times 1000}{3600} = 9.72 \text{ m/s}$$

$$u_2 = \frac{60 \times 1000}{3600} = \underline{\underline{16.67 \text{ m/s}}}$$

$$\text{Radius of the wheel } r = \frac{0.6}{2} = \underline{\underline{0.3 \text{ m}}}$$

$$\text{Time taken, } t = 8 \text{ sec.}$$

Let find the corresponding initial and final angular velocities ω_1 and ω_2 respectively using the corresponding initial and final linear velocities and radius of the wheel.

$$\Rightarrow v = \omega r$$

$$u = \omega_1 r \Rightarrow \omega_1 = \frac{u}{r} = \frac{9.72}{0.3} = \underline{\underline{32.4 \text{ rad/s}}}$$

$$\omega_2 = \frac{v}{r} = \frac{16.67}{0.3} = \underline{\underline{55.6 \text{ rad/s}}}$$

(a) The angular acceleration of the wheel, $\alpha = \frac{\omega_2 - \omega_1}{t}$

$$= \frac{55.6 - 32.4}{8}$$

$$\therefore \alpha = \underline{\underline{2.9 \text{ rad/sec}^2}}$$

(b) During the acceleration, the angular velocity changes from ω_1 to ω_2 in time t .

$\therefore$ the angular displacement θ during this acceleration is given by: $\theta = \frac{1}{2} (\omega_1 + \omega_2) t$

$$= \frac{1}{2} (32.4 + 55.6) \times 8$$

$$\therefore \theta = 352 \text{ rad}$$

There is 2π rad for every revolution

$$\Rightarrow 2\pi \text{ rad} = 1 \text{ revolution}$$

$$352 \text{ rad} = \frac{1 \text{ rev} \times 352 \text{ rad}}{2\pi \text{ rad}} = \underline{\underline{56 \text{ revolution}}}$$

(4) A flywheel increases its speed from 30 rev/min to 60 rev/min in 10 sec. The diameter of the wheel is 350 mm. Cal.

(a) The angular acceleration

(b) The no. of revolution made during the 10 sec.

(c) The linear acceleration of a Pt. on the rim of the wheel.

Let the initial and final speeds be N_1 and N_2 respectively. (12)
 Now convert the speeds to rad/sec.
 Diameter of the flywheel, $D = 3500 \text{ mm} = 3.5 \text{ m}$
 time taken, $t = 10 \text{ sec}$.

$$\text{i.e., } \omega_1 = \frac{2\pi N_1}{60}$$

$$= \frac{2 \times \pi \times 30}{60} = 3.14 \text{ rad/sec}$$

Similarly,

$$\omega_2 = \frac{2\pi N_2}{60} = \frac{2 \times \pi \times 60}{60} = 6.28 \text{ rad/s}$$

(a) Hence, angular acceleration $\alpha = \frac{\omega_2 - \omega_1}{t}$

$$= \frac{6.28 - 3.14}{10}$$

$$\therefore \alpha = 0.314 \text{ rad/s}^2$$

(b) First, determine the angular displacement θ in radians during the same period.

$$\text{Angular displacement } \theta = \frac{1}{2}(\omega_1 + \omega_2)t$$

$$= \frac{1}{2}(3.14 + 6.28) \times 10$$

$$\therefore \theta = 47.1 \text{ rad}$$

There is 2π rad for every revolution.
 $\Rightarrow 2\pi \text{ rad} = 1 \text{ revolution}$

$$47.1 \text{ rad} = 1 \text{ rev} \times 47.1 \text{ rad}$$

$$= 7.5 \text{ revolution}$$

(c) To determine the linear acceleration of a pt. of the flywheel

$$a = r\alpha$$

$$\text{but } r = \frac{D}{2} = \frac{3.5}{2} = 1.75 \text{ m}$$

$$= 1.75 \times 0.314$$

$$a = 0.55 \text{ m/s}^2$$

(13)

5. A vehicle with wheels 700mm diameter has its speed reduced from 50 km/h to 30 km/h while it travels a distance of 22m. cal.

- The time taken
- The linear retardation of the vehicle
- The angular velocity of the wheels at 50 km/h
- The angular retardation of the wheels

Solution:

$$\text{Radius of wheel, } r = \frac{0.7}{2} = 0.45\text{m}$$

$$\text{Initial linear velocity, } u = 50\text{ km/h}$$

$$= \frac{50 \times 1000}{3600} = 13.89\text{ m/s}$$

$$\text{Final linear velocity, } v = \frac{30 \times 1000}{3600} = 8.33\text{ m/s}$$

$$\text{Distance travel, } s = 22\text{m.}$$

a) The time taken can be determined from the relation
 $s = \frac{1}{2}(u+v)t$

$$22 = \frac{1}{2}(13.89 + 8.33) \times t$$

$$\therefore t = \underline{1.98\text{ sec}}$$

b) The linear retardation, a , can be determined from the relation: $v = u + at$

$$8.33 = 13.89 + a \times 1.98$$

$$\therefore a = \underline{-2.8\text{ m/s}^2}$$

c) angular velocity ω_1 of the wheels at 50 km/h (13.89 m/s)

$$\Rightarrow \omega_1 = \frac{u}{r} = \frac{13.89}{0.45} = \underline{30.8\text{ rad/s}}$$

d) Final angular velocity ω_2 of the wheels at 30 km/h (8.33 m/s)

$$\omega_2 = \frac{v}{r} = \frac{8.33}{0.45} = \underline{18.5\text{ rad/sec}}$$

The angular retardation of the wheels α and determine as:

(14)

$$\alpha = \frac{w_2 - w_1}{t} = \frac{18.5 - 30.8}{1.98} = \underline{\underline{-6.2 \text{ rad/s}^2}}$$

$\therefore$ The angular retardation of the wheels is 6.2 rad/s^2

(6) A 600mm diameter Pulley lifts a load at 0.3 m/s initially, and over a Period of 10sec, the speed is uniformly accelerated to 1.4 m/s : cal:

(a) The acceleration of the load:

(b) The angular acceleration of the load.

(c) The distance moved by the Pulley

(d) The number of turns made by the load.

Solution

$u = 0.3 \text{ m/s}$, $v = 1.4 \text{ m/s}$, $t = 10 \text{ sec}$.

radius of the Pulley, $r = \frac{0.6}{2} = \underline{\underline{0.3 \text{ m}}}$

(a) acceleration of the load: $v = u + at$

$$1.4 = 0.3 + a \times 10$$

$$a = \underline{\underline{0.11 \text{ m/s}^2}}$$

(b) angular acceleration α , of the Pulley:

$$a = r\alpha$$

$$\Rightarrow \alpha = \frac{a}{r} = \frac{0.11}{0.3} = \underline{\underline{0.36 \text{ rad/s}^2}}$$

(c) Distance moved by the load can be determined as:

$$s = \frac{1}{2} (u + v) t$$

$$= \frac{1}{2} (0.3 + 1.4) \times 10 = \underline{\underline{8.5 \text{ m}}}$$

(d) Number of turns made by the Pulley, first find the initial and final angular velocities w_1 and w_2 respectively

$$\omega_1 = \frac{v}{r} = \frac{0.3}{0.3} = 1 \text{ rad/sec}$$

$$\omega_2 = \frac{v}{r} = \frac{1.4}{0.3} = 4.67 \text{ rad/s}$$

The angular displacement θ of the pulley can be determined as $\theta = \frac{1}{2}(\omega_1 + \omega_2)t$

$$= \frac{1}{2}(1 + 4.67) \times 10$$

$$\therefore \theta = 28.33 \text{ rad}$$

There is 2π rad of every revolution

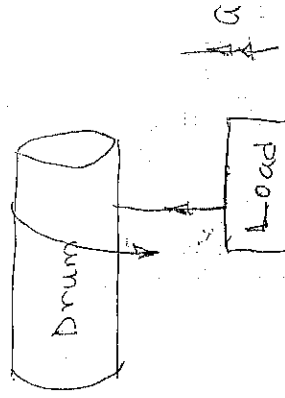
$$\Rightarrow 2\pi \text{ rad} = 1 \text{ revolution}$$

$$\therefore 28.33 \text{ rad} = \frac{28.33 \text{ rad} \times 1 \text{ revolution}}{2\pi \text{ rad}} = 4.5 \text{ revolutions}$$

7. A load is lifted by a cable which is wound round a drum of 180 mm diameter. The drum is initially running at 20 rev/min and is uniformly accelerated to 40 rev/min in 6 sec. Determine:

- The initial velocity of the load in m/s
- The angular acceleration of the drum in rad/s^2
- The acceleration of the load in m/s^2
- The distance in m moved by the load.

Solution:



$$N_1 = 20 \text{ rev/min}$$

$$N_2 = 40 \text{ rev/min}$$

$$t = 6 \text{ sec.}$$

$$\text{radius } (r) = \frac{0.18}{2} = 0.09 \text{ m}$$

$$\text{a) The initial angular speed } \omega_1 = \frac{2\pi N_1}{60} = \frac{2\pi \times 20}{60}$$

$$= 2.09 \text{ rad/s}$$

∴ The initial linear speed $u = r\omega_1$

(16)

$$\therefore u = 0.09 \times 2.09 = \underline{\underline{0.188 \text{ m/s}}}$$

(b) The final angular speed ω_2 of the drum

$$\begin{aligned} \omega_2 &= \frac{2\pi N_2}{60} \\ &= \frac{2 \times \pi \times 40}{60} = \underline{\underline{4.19 \text{ rad/s}}} \end{aligned}$$

The angular acceleration of the drum α is given by

$$\begin{aligned} \alpha &= \frac{\omega_2 - \omega_1}{t} \\ \therefore \alpha &= \frac{4.19 - 2.09}{6} = \underline{\underline{0.35 \text{ rad/s}^2}} \end{aligned}$$

(c) The acceleration of the load $a = r\alpha$

$$\begin{aligned} &= 0.09 \times 0.35 \\ a &= \underline{\underline{0.0315 \text{ m/s}^2}} \end{aligned}$$

(d) The distance moved by the load can be determined by:

$$\begin{aligned} s &= ut + \frac{1}{2}at^2 \\ &= 0.188 \times 6 + \frac{1}{2} \times 0.035 \times (6)^2 \\ s &= \underline{\underline{1.69 \text{ m}}} \end{aligned}$$

Circular Motion: — is defined as the movement of an object along the circumference of a circle or rotation along a circular path. Circular motion can be uniform or non-uniform. It is said to be uniform when an object moves along the circumference of a circle with constant speed and constant angular rate of rotation and it is non-uniform when it is characterised by changing rate of rotation.

In a uniform circular motion, an object moves round a circle with a constant speed but the acceleration keeps in

changing b/c of change in directions. The direction of this acceleration is always directed towards the center of the circle.

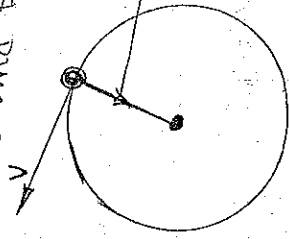
(17)

According to Newton's 2nd law of motion, acceleration of an object brought about by the net force acting on that object. $\therefore$, towards the center of the circle the inward force is directed. Inward force is responsible for keeping the object moving in circular path.

Under this condition, the inward force is called the Centripetal force while the acceleration is called Centripetal acceleration. It is important to note that when an object is moving round a circular path the speed is constant but the vel. as well as the acceleration changes from point to point.

Centripetal Acceleration and Centripetal Force:

When an object is moving in circular path with a constant speed, as stated earlier, the acceleration of the object keeps changing b/c of the changes in the direction of the object at any instant. This acceleration that is always directed towards the center of a circle is called Centripetal acceleration while the force pulling the object towards the center of the circle is called Centripetal force. $\therefore$, Centripetal force is a function of the Centripetal acceleration and both are directed towards the center of the circle.



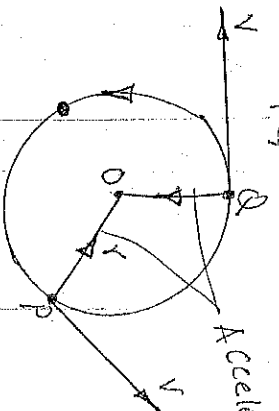
Centrifugal Acceleration and Centrifugal Force:
When an object is moving in a circular path, the outward force exerted on the body is called the centrifugal force.

(18) The acceleration produced as a result of the outward force on the body is called the centrifugal acceleration.

Expressions for Centripetal and Centrifugal acceleration.

Consider a heavy object attached to the end of a rope and whirled steadily round in a horizontal circle by holding the

other end of the rope;



This object moves with a constant speed round the circle but the velocity v changes from point to point, this velocity is always directed along the tangent to the circle at the point concerned. Any object moving in a circle requires a force directed towards the center to keep it from moving along the tangent path. This force on the object is called the centripetal force. Since a force produces acceleration, this acceleration of the object must be directed towards center O .

Hence, centripetal force $F_c = Ma$

where $M =$ mass of the object moving round the circular path.
 $a =$ centripetal acceleration.

But $a = \omega^2 r$

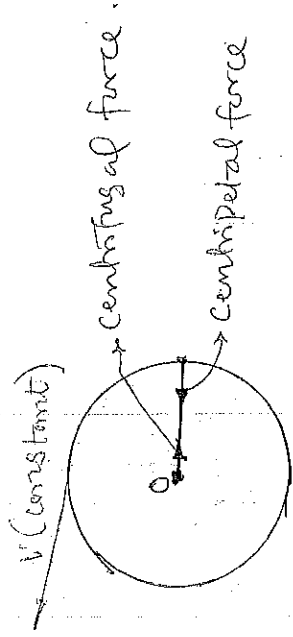
where $\omega =$ angular velocity
 $r =$ radius of the circular path.

$\Rightarrow F_c = m\omega^2 r$ Since the centripetal force acts radially,

According to Newton's third law of motion, the object must exert a force radially outwards of equal magnitude. This force is called the centrifugal force, F_{cf} , and is given by

$$F_{cf} = m\omega^2 r$$

(19)



Centrifugal Effects:

Examples of centrifugal effects are:

- (i) Movement of a car round a circular path: The centrifugal force provided by the friction b/w the tyres and the road helps to maintain a car on a circular track. This is so b/c, if the frictional force is not enough to provide the necessary centrifugal force, the car will move outwards or skid.
- (ii) Planetary motion
- (iii) Conical pendulum.

Worked Example:

1) The crankpin of an engine has a mass of 0.8 kg and its center of gravity is 135 mm from the crankshaft center. When the crankshaft is rotating uniformly at 1500 rev/min , calculate the magnitude and direction in each case:

- a) The acceleration of the crankpin
- b) The centripetal force on the crankpin
- c) The centrifugal force on the crankpin

Solution:

Mass of crankpin, $m = 0.8 \text{ kg}$

Speed of rotation of crankpin, $N = 1500 \text{ rev/min}$

The distance b/w the crankpin and the crankshaft center is the radius of the turn $r = 0.135 \text{ m}$

The angular speed of the crankshaft ω , can be determined from the relation:

$$\omega = \frac{2\pi N}{60} = \frac{2\pi \times 1500}{60} = 157.1 \text{ rad/s}$$

Momentum of Bodies:

Momentum of a body is defined as the product of its mass and its velocity.

(22)

Mathematically,

$$\text{Momentum} = \text{Mass (m)} \times \text{velocity (v)} \\ = mv$$

Units of Momentum is kgm/s.

Worked Example:

1. Find the momentum of the body in each of the following:

(a) Mass 2kg, velocity, 5m/s

(b) Mass

0.25kg, velocity, 2m/s

(c) Mass

300kg, velocity, 20km/h

Solution:

$$(a) \text{ Momentum} = \text{Mass} \times \text{velocity} \\ = 2 \times 5 = \underline{\underline{10 \text{ kgm/s}}}$$

$$(b) \text{ Momentum} = \text{Mass} \times \text{velocity} \\ = 0.25 \times 2 = \underline{\underline{0.5 \text{ kgm/s}}}$$

$$(c) \text{ Momentum} = \text{Mass} \times \text{velocity} \\ = 300 \times \frac{20 \times 1000}{3600}$$

$$= \underline{\underline{1667 \text{ kgm/s}}}$$

(2) A Peugeot car of mass 1500kg, moving with a constant acceleration of 8 m/s^2 is acted upon by a force exerted from the engine to maintain the forward acceleration.

(20)
 a) Radius of the crankpin $r = 0.135\text{m}$, then Centripetal ~~for~~ acceleration a_c of the crankpin can be determined from the relation: $a = \omega^2 r$

$$= (157.1)^2 \times 0.135$$

$$= 3332 \text{ m/s}^2 \text{ (toward the center of turn)}$$

b) Centripetal force on the crankpin (F_{cp}) can be determined from the relation: $F_{cp} = m\omega^2 r$

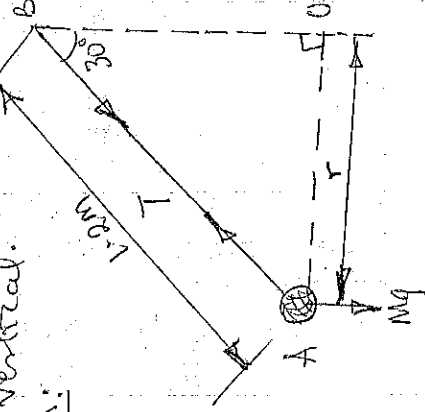
$$= 0.8 \times (157.1)^2 \times 0.135$$

$$F_{cp} = \underline{2665 \text{ N}}$$

c) Since F_{cp} is equal to F_{cf} in magnitude but opposite in direction, $\therefore F_{cf}$ on the crankshaft bearing is 2665 N acting radially outwards from the centre of turn.

2) An object of mass m , is hung at the end of an elastic rope of length 1.2m . Determine the speed in rev/min at which the object needs to rotate to keep the rope at an angle of 30° to the vertical.

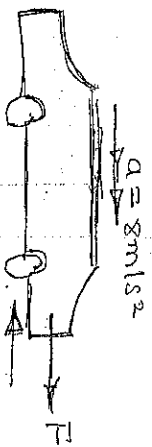
Solution:



The Centripetal force, F_{cp} , should be determined wrt the horizontal circle of radius r , $\therefore$ the centripetal force can be found by resolving the tension T into vertical and horizontal components.

$$\sin 30 = \frac{r}{L} = \frac{r_{hyp}}{L_{hyp}} \quad \cos 30 = \frac{Adj}{hyp} =$$

$$r = T \sin 30^\circ \quad T \cos 30^\circ$$



Let F = force of the engine
 F_r = Resistive force = 1200 N

a = acceleration of the car = 8 m/s^2

M = Mass of the car = 1500 kg

Resultant force acting on the car = $F - F_r$

The resultant force ($F - 1200$) acting on the car is
 equal to the acceleration force (Ma)

$$\Rightarrow F - 1200 = Ma$$

$$\begin{aligned}\Rightarrow F &= Ma + 1200 \\ &= (1500 \times 8) + 1200 \\ &= 10,800 \text{ N}\end{aligned}$$

Impulse of a force:

Impulse of a force is defined as the Product of a large force and a very short time during which it acts. Also, impulse can be defined as the time effect of the force.

The impulse of a force = Ft

The unit of impulse is Newton second (Ns)

$$F = \frac{m(v-u)}{t}$$

$$Ft = m(v-u)$$

$\therefore$ Impulse of a force (Ft)

Note! Since the impulse of a force is equal to the change in momentum, $m(v-u)$, momentum can also be

expressed in Ns.

(24)

Worked Example:

- 1 A body of mass 10kg moving with a speed of 40m/s is suddenly hit by another body moving in the same direction, thereby changing the speed of the former body to 70m/s. What is the impulse received by the first body?

Solution:

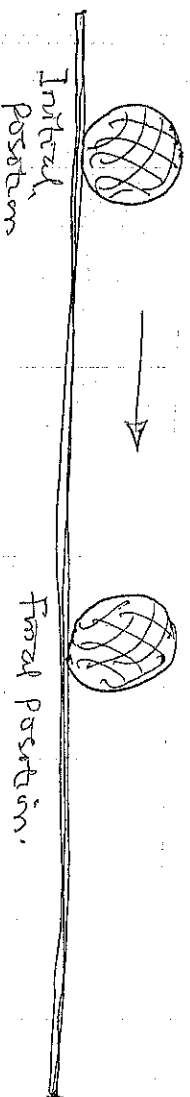
Let the information be represented below:

$$M = 10\text{kg}$$

$$u = 40\text{m/s}$$

$$M = 10\text{kg}$$

$$v = 70\text{m/s}$$

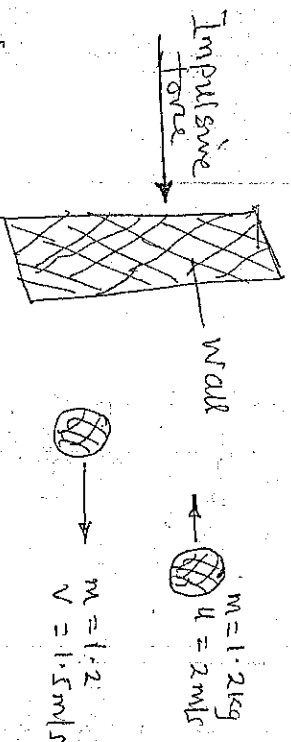


$$\text{Impulse} = \text{change in momentum}$$

$$= M(v - u)$$

$$= 10(70 - 40) = \underline{300 \text{ Ns}}$$

- 2 A mass of 1.2kg travelling at 2m/s strikes a wall at right angle and rebounds (also at right angle) with a velocity of 1.5m/s. If the contact lasted 0.25sec, find the force on the mass.



$$\text{Impulse} = \text{change in momentum}$$

$$\Rightarrow \text{Impulse} = \text{Final momentum} - \text{Initial momentum}$$

$$Ft = Mv - (-Mu)$$

$$= (1.2 \times 1.5) - (-1.2 \times 2)$$

$$\therefore Ft = 4.2 \text{ Ns}$$

Since the contact lasted for 0.25s

It implies that

$$Ft = 4.2$$

$$F \times 0.25 = 4.2$$

$$\therefore \underline{F = 16.8 \text{ N}}$$

(25)

③ A hammer of mass 5kg travelling at 4m/s, hits a nail directly and does not rebound.

① what is the impulse on the hammer.

② If the contact effectively lasts 0.5sec, what is the average force b/w the two.

Solution:

$$m = 5 \text{ kg}$$

$$u = 4 \text{ m/s}$$

Since the hammer does not rebound, $\Rightarrow mv = 0$
but, Impulse = change in momentum

$$= \text{Final momentum} - \text{Initial momentum}$$

$$Ft = mv - (-mu)$$

$$= (5 \times 0) - (5 \times 4)$$

$$Ft = \underline{20 \text{ Ns}}$$

$$\Rightarrow t = 0.5$$

$$= F \times 0.5 = 20$$

$$\text{So that } F = \underline{40 \text{ N}}$$

Assignment

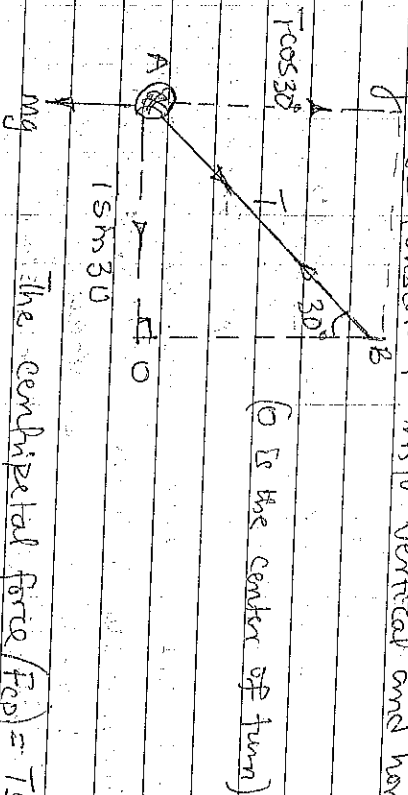
① A ball of mass 0.5kg falls freely onto a concrete floor from a height of 25m, and rebounds to a vertical height of 7m. If the ball was in contact with the floor for 0.95sec, find the average force exerted on the ball

[Take $g = 9.8 \text{ m/s}^2$]

② A horizontal jet of water is emitted from a circular pipe of radius 100mm at speed of 15m/s. Find the mass of water emitted per second and the average force exerted on a vertical wall.

(21)

by resolving the tension T into vertical and horizontal components



The centripetal force (F_{cp}) = $T \sin 30$

$$\Rightarrow F_{cp} = T \sin 30 \quad \text{--- (i)}$$

But Centripetal force F_{cp} can be expressed as

$$F_{cp} = m\omega^2 r \quad \text{--- (ii)}$$

Equate eqn (i) and (ii)

$$T \sin 30 = m\omega^2 r$$

$$\Rightarrow 0.5T = m\omega^2 r \quad \text{--- (ii)}$$

From ΔAOB ,

$$\sin 30 = \frac{r}{1.2}$$

$$r = 1.2 \sin 30$$

$$r = 0.6 \text{ m}$$

Now substitute r in equation (ii) above

$$\Rightarrow 0.5T = m\omega^2 \times 0.6 \quad \text{--- (iv)}$$

Since the object has not displaced vertically, $\therefore$ the force must be in equilibrium

$$\Rightarrow T \cos 30 = mg$$

$$T = \frac{mg}{\cos 30} \quad \text{--- (v)}$$

$$\text{Take } g = 9.8 \text{ m/s}^2$$

Substitute eqn (v) in eqn (iv)

$$0.5 \times \frac{m \times 9.8}{\cos 30} = m \times \omega^2 \times 0.6$$

$$\frac{\sqrt{3}}{2}$$

$$\frac{2}{\sqrt{3}}$$

$$\Rightarrow 0.5 \times \frac{m \times 9.8}{\sqrt{3}} \times 2 = m \times \omega^2 \times 0.6$$

$$\frac{\sqrt{3}}{2}$$

$$\omega^2 \times 0.6 \times \sqrt{3} = 0.5 \times 9.8 \times 2$$

$$\omega^2 = \frac{0.5 \times 9.8 \times 2}{0.6 \times \sqrt{3}}$$

$$\therefore \omega = 3.96 \text{ rad/s}$$

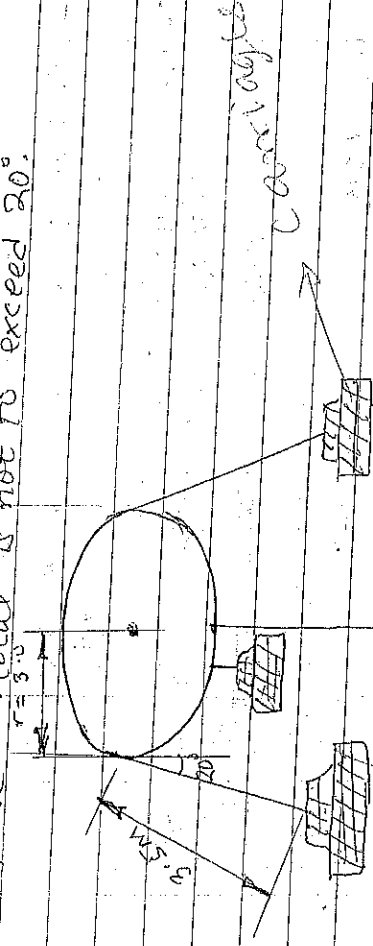
(72)

The speed N of the object in rev/min can be determined
 $\omega = \frac{2\pi N}{60}$

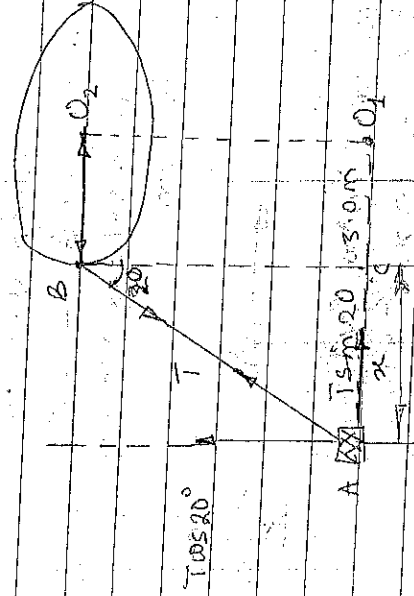
$$N = \frac{\omega \times 60}{2\pi} = \frac{3.96 \times 60}{2\pi}$$

$$N = 37.8 \text{ rev/min}$$

(3) A roundabout has a number of carriages hung from a rotatory roof by means of chains 3.5m long. The upper ends of the chains are attached to the circular roof of 3.0m radius. Find the greatest speed in rev/min at which the roundabout can operate, if the angle which the chains make with the vertical is not to exceed 20° .



The horizontal and vertical components of the tension, T



Centripetal force (F_{cp}) is equal to $T \sin 20^\circ$, this implies that

$$F_{cp} = T \sin 20^\circ \quad \text{--- (i)}$$

But centripetal force $F_{cp} = m\omega^2 r$

$$\text{Hence, } T \sin 20^\circ = m\omega^2 r \quad \text{--- (ii)}$$

(2.3)

where r is the radius of turn $= AO_1$

$$\Rightarrow AO_1 = AC + CO_1$$

$$AO_1 \Rightarrow x + 3$$

$$\text{but } CO_1 = BO_2 = 3$$

$$\text{From } \Delta ACB, \sin 20^\circ = \frac{x}{3.5} \Rightarrow x = 3.5 \sin 20^\circ$$

$$\therefore \text{radius of turn, } AO_1 = 1.197 + 3 = \underline{\underline{4.197 \text{ m}}}$$

$$\therefore r = 4.197 \text{ m (since } r = AO_1)$$

Form equation (ii)

$$T \sin 20^\circ = M \times \omega^2 \times 4.197 \quad \text{--- (ii)}$$

Since there is no vertical displacement of the carriage, the vertical forces must be in equilibrium

$$T \cos 20^\circ = Mg$$

$$T = \frac{Mg}{\cos 20^\circ} = \text{--- (iv)}$$

$$Mg \sin 20^\circ = M \times \omega^2 \times 4.197$$

$$\omega \tan 20^\circ = \omega^2 \times 4.197$$

$$\omega^2 = \frac{g \tan 20^\circ}{4.197}$$

$$\text{but } g = 9.81 \text{ m/s}^2$$

$$\omega = \frac{9.81 \times 0.364}{4.197} = \underline{\underline{0.922 \text{ rad/s}}}$$

The speed N in rev/min at which the car should operate

$$\omega = 2\pi N \Rightarrow N = \frac{0.922 \times 60}{2\pi}$$

$$N = \frac{0.922 \times 60}{2 \times 3.142} = \underline{\underline{8.8 \text{ rev/min}}}$$

② Distance travelled during deceleration =

⑦

Area of ΔBCE

$$= \frac{1}{2} \times 10 \times 30 = \underline{\underline{150m}}$$

(f) Total distance travelled = Area of trapezium OABC

$$= \frac{1}{2} (55 + 30) \times 30$$

$$= \underline{\underline{1275m}}$$

or Alternatively, total distance travelled can be found by the summation of the area of $\Delta AOD + \square ABED +$

$$\Delta BCE = 225 + 900 + 150 \\ = \underline{\underline{1275m}}$$

(g) The deceleration is the slope on line BC

$$= \frac{30 - 0}{45 - 55} = \underline{\underline{-3}}$$

Note that, the deceleration is $\underline{\underline{3m/s^2}}$, the value must be + sign.

③ A bus takes 3 minutes to travel b/w two station 3100m apart. The bus accelerates uniformly to a speed of 20m/s and the travel for a time at this speed b/w decelerating uniformly to rest at the second station.

If the acceleration and deceleration are the in the ratio 2:3 Calculate.

- (a) The time for the bus to travel at uniform speed.
- (b) The time taken during the acceleration.
- (c) Its acceleration.

$$V = u + at$$

$$S = ut + \frac{1}{2}at^2$$

$$V^2 = u^2 + 2as$$

$$V^2 = u^2 + 2as$$

$$= 2s^2 + 2(-1) \times 150$$

$$V^2 = 400 - 300$$

$$V = \sqrt{100}$$

$$V = \sqrt{10 \text{ m/s}}$$

$$x = 8t$$

$$x = 120 - x$$

$$x - S = \frac{1}{2}(u+v)t$$

$$= \frac{1}{2}(16+0) \times t$$

$$\therefore x = 8t$$

$$a = \left(\frac{-16}{8.57} \right) \text{ m/s}^2$$

$$= -1.87 \text{ m/s}^2$$

$$V = \text{displacement}$$

$$t = \text{time}$$

$$= \text{m/s}$$

$$V = \frac{s}{t}$$

$$a = \frac{V}{t}$$

$$\text{from (i)}$$

$$t = \frac{s}{V}$$

$$a = \frac{V}{\frac{s}{V}} = \frac{V^2}{s}$$

$$\text{putting (ii) in (i)}$$

$$a = \frac{V}{\frac{s}{V}} = \frac{V^2}{s}$$

$$\Rightarrow \boxed{a = \frac{V^2}{s}}$$

$$\text{Area of trapezium} = \frac{1}{2}(AD+BC) \times h$$

$$V = \frac{s}{t}$$

$$a = \frac{V}{t}$$

$$V^2 = u^2 + 2as$$

$$V^2 = 400 - 300$$

$$V = \sqrt{100}$$

$$V = \sqrt{10 \text{ m/s}}$$

$$x = 8t$$

$$x = 120 - x$$

$$x - S = \frac{1}{2}(u+v)t$$

$$= \frac{1}{2}(16+0) \times t$$

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$$= -1.87 \text{ m/s}^2$$

$$V = \text{displacement}$$

$$t = \text{time}$$

$$= \text{m/s}$$

$$V = \frac{s}{t}$$

$$a = \frac{V}{t}$$

$$\text{from (i)}$$

$$t = \frac{s}{V}$$

$$a = \frac{V}{\frac{s}{V}} = \frac{V^2}{s}$$

$$\text{putting (ii) in (i)}$$

$$a = \frac{V}{\frac{s}{V}} = \frac{V^2}{s}$$

$$\Rightarrow \boxed{a = \frac{V^2}{s}}$$

$$s = \frac{1}{2}(u+v)t$$

$$\text{multiply through by 2}$$

$$2s = (u+v)t$$

$$2s = ut + vt$$

$$V = u + at$$

$$V - u = at$$

$$a = \frac{V - u}{t} = \frac{0 - 16}{t}$$

$$a = \left(\frac{-16}{t} \right) \text{ m/s}^2$$

$$S = \frac{1}{2}(u+v)t$$

$$a = \frac{V - u}{t}$$

$$a = \left(\frac{-12}{t} \right) \text{ m/s}^2$$

$$120 - x = \frac{1}{2}(12 + 0) \times t$$

$$120 - x = 6t$$

$$120 - 8t = 6t$$

$$120 = 6t + 8t$$

$$120 = 14t$$

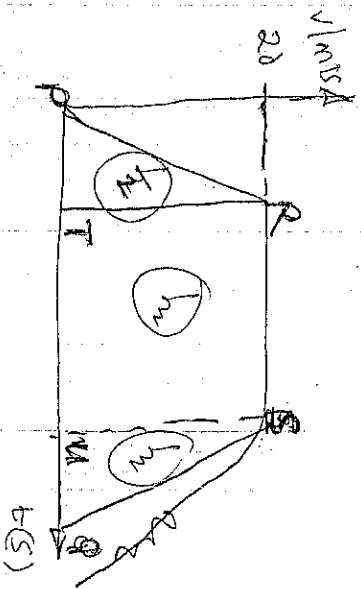
$$t = \frac{120}{14} = 8.57s$$

Never

2nd

34138313332311745

293 942 0374 6418



(3)

(a) The total distance travelled by bus, S, is equal to area of trapezium PQRS

$$\therefore S = \frac{1}{2} (RS + PQ) \times 200 \text{ (m)}$$

The total time taken for the bus to travel from P to Q is 3 minutes i.e. $30 \times 60 = 180 \text{ sec}$ (m)

$$= 3100 = \frac{1}{2} (RS + 180) \times 20 \text{ (m)}$$

$$RS = 130 \text{ sec. (m)}$$

(b) Let the acceleration be a and the deceleration be

$$d \therefore a = \frac{RT}{PT} \text{ and } d = \frac{SQ}{UQ} \text{ (m)}$$

$$a \text{ and } d = \frac{RT}{PT} \times \frac{SQ}{UQ} = 2:3 \text{ (m)}$$

$$\frac{20}{PT} \times \frac{20}{UQ} = \frac{2}{3} \text{ (m)}$$

$$\frac{20}{PT} \times \frac{UQ}{20} = \frac{2}{3} \text{ (m)}$$

$$UQ = \frac{2PT}{3} \Rightarrow UQ = \frac{2PT}{3} \text{ (m)}$$

Since the total time taken for the bus to travel from P to Q is 3 minutes (180 sec), it implies that

$$PT + TU + UQ = 180 \text{ (m)}$$

$$\text{But } TU = RS = 130 \text{ and } UQ = \frac{2PT}{3} \text{ (m)}$$

$$\Rightarrow PT = 130 + \frac{2PT}{3} \text{ (m)}$$

$$PT = 30 \text{ seconds (m)}$$

(c) To find the acceleration, a

$$a = \frac{RT}{PT} \text{ (m)}$$

$$= \frac{20}{30} \text{ (m)}$$

$$\therefore a = 0.66 \text{ m/s}^2 \text{ (m)}$$

But $TU = RS = 130$ and $UQ = \frac{2PT}{3}$ (m)

$$w(t + T) + \phi = (wt + \phi) + 2\pi$$

Open the bracket

$$wt + wT + \phi = (wt + \phi) + 2\pi$$

Re-arrange

$$(wt + \phi) + wT = (wt + \phi) + 2\pi$$

Subtract $(wt + \phi)$ from the L.H.S and R.H.S yields

$$wT = 2\pi$$

$$\Rightarrow T = \frac{2\pi}{w}$$

The Period T , and the angular frequency (w) are inversely related, meaning that the smaller the angular frequency the larger the Period and the more slowly the body completes a cycle.

Conversely, the larger the angular frequency, the smaller the Period and the more quickly the body completes a cycle.

The frequency f is a quantity that is used to specify the rate of oscillation or vibration of a body and is inverse of the Period T . This implies that:

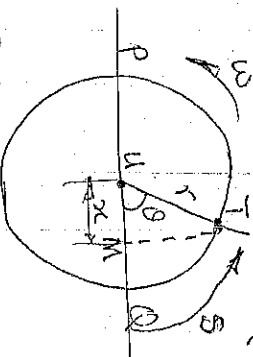
$$f = \frac{1}{T}$$

The Period T is the time per cycle. Therefore, the frequency f , is the number of cycles per unit time. Relationship b/w S.H.M and Circular Motion:

To establish a relationship b/w simple harmonic motion and circular motion, let point T be a point moving round a circle of radius r , with uniform speed v . As T moves round the circle, at any instant a perpendicular line can be drawn from T to meet line PQ at m .

distance PQ is the diameter of the circle.

A little consideration will show that as T moves round circle once, the foot of the Perpendicular M moves from Q through U to P and back to Q. (45)



Therefore, as Point T moves round the circle, Point M will move along PQ with simple harmonic motion (SHM). The angular velocity ω of the Point T is defined as:

$\omega = \frac{\text{angle turned through by T}}{\text{time taken.}}$

$$\Rightarrow \omega = \frac{\theta}{t} \Rightarrow \theta = \omega t$$

But $S = r\theta$ it implies that $S = r\omega t$

It can be seen that the corresponding linear velocity V , is given by:

$$V = \frac{S}{t}$$

where S is the length of the arc b/w Q and T at time t . The length of the arc S is given by: $S = r\theta$ as shown in eqn. (i) above, where θ is expressed in radians. Worked Example:

A body makes 120 rpm on a circle of 3m radius. Find its

(i) Period (ii) Angular Velocity (iii) Linear velocity and (iv) Linear acceleration.

Solution:

(i) Radius of circle, $r = 3\text{m}$
frequency, $f = 120\text{rpm}$

$$= \frac{120}{60} = 2 \text{ rev. per second.}$$

Period T of oscillation is given by:

$$T = \frac{1}{f} = \frac{1}{2} = 0.5 \text{ sec} \quad \therefore T = 0.5 \text{ sec}$$

Simple Harmonic Motion: Periodic Motion:-

Chapter six:

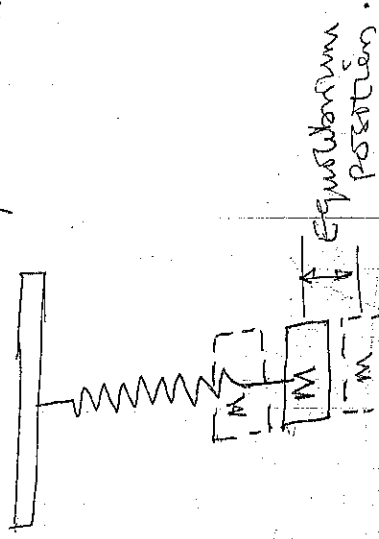
42

Simple harmonic motion is a type of periodic motion where the restoring force is directly proportional to the displacement. This periodic motion is typified by the motion of a mass on a spring when it is subjected to the linear elastic restoring force.

When Hooke's law is applied to the mass-spring system, the restoring force F is given by:

$$[F = -Kx]$$

where F is the restoring elastic force exerted by the spring, K is the spring constant and x is the displacement from the equilibrium position.



It can be seen that once the mass is displaced from its equilibrium position, it experiences a net restoring force. Because of this, the mass accelerates and starts going back to the equilibrium position. The restoring force decreases as the mass moves

towards the equilibrium position. At the equilibrium position, the restoring force is zero.

Therefore, the simple harmonic motion is a periodic type of motion.

Examples of Periodic motion are:

- (i) Seasons of the year
- (ii) The heartbeat of an animal
- (iii) The vibration of atoms in solids
- (iv) The swinging of a pendulum e-t.c.

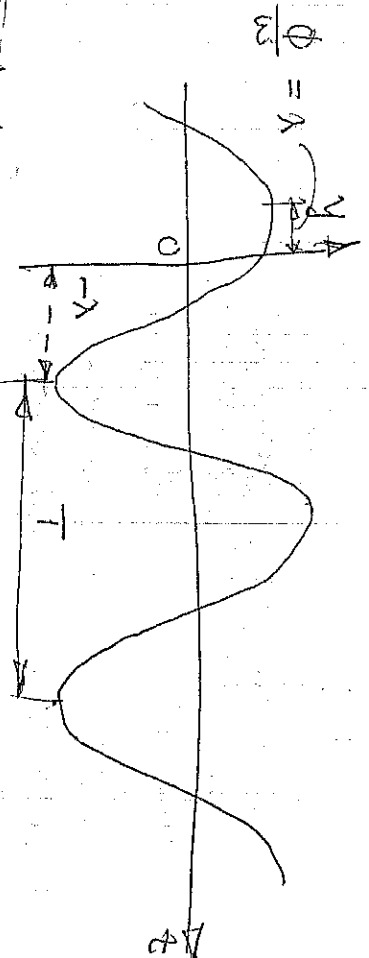
The variation of the mass coordinate sinusoidally is called S.H.M, if y is the coordinate of the mass m undergoing S.H.M then,

$$y = A \sin(\omega t + \phi)$$

When the body oscillates one way or the other, then y will vary sinusoidally with time b/w $y = A$ and $y = -A$.

The maximum displacement of the mass is called the Amplitude (A) while ω denotes the angular frequency whose values determines the rate of the oscillations of the mass.

The Phase Constant ϕ is selected at an instant when $t = 0$.



In simple harmonic motion, the motion of the mass m repeats itself after a characteristic time interval T . Units of time interval is called the Period of the S.H.M. Little consideration will show that for a full cycle, the phase ($\omega t + \phi$) increases by 2π while the time interval t increases by T , this implies that:

Simple Machines:

Chapter Five

(38)

A simple machine is defined as a contrivance which enables small force (effort) acting at a point to overcome large force (Load) acting at some other point.

Example of simple machines are: Pulley system

screw Jack

gear system

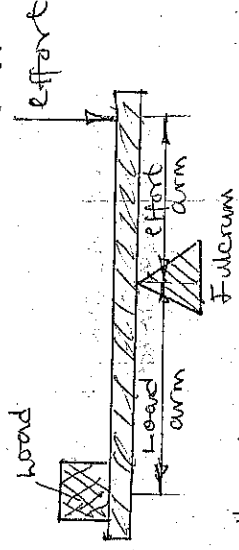
and lever e-t-c

A lever is a simple machine that is used for lifting a heavy load on one end of the lever by applying a small effort on the other end. In fact heavy loads that are beyond our strength limit can easily be lifted with the use of levers.

There are basically three classes of levers, class 1, class 2 and class 3. These classes are classified according to the location of the pivot or fulcrum.

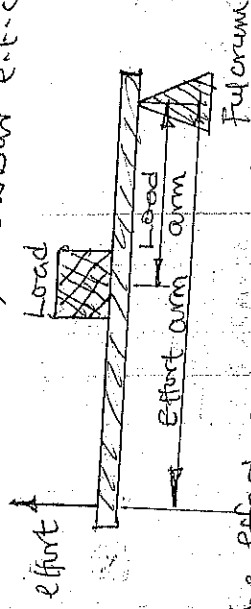
Class 1: - The fulcrum is located b/w the effort and the load as shown below. Examples of this class of lever are:

Pair of pliers, scissors, catapult e-t-c.



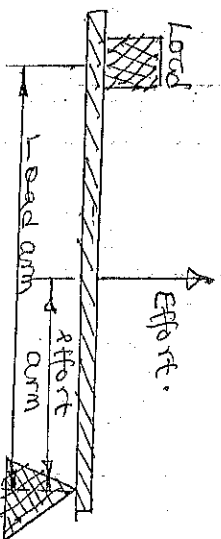
Class 2: - Here, the load is located b/w the effort and the fulcrum as shown. Examples of this class of lever are:

Nut cracker, wheelbarrow, crowbar e-t-c.



Class 3: - The effort is located b/w the load and the fulcrum as shown, examples of this class of lever are:

Stapler, hockey stick and mouse trap e-t-c.



Fulcrum

Pulleys - one simple machines used for lifting heavy load to higher elevation. Examples where pulleys are applied in docks where cranes are used to lift heavy cranes in and out of the ships.

In pulley system, a continuous rope is passed round the pulleys where the weight to be lifted is attached to one end of the rope and the effort is applied at the other end.

A Screw Jack - is a simple machine that is used for lifting heavy loads such as cars. Since the screw thread has threads, the load is raised through a distance equal to the pitch (for single-start thread) by the effort through one revolution.

Mechanical Advantage - (M.A) is defined as the ratio of load to effort.

$$\text{Mathematically, } M.A = \frac{\text{Load (L)}}{\text{Effort (E)}} = \frac{W}{E}$$

Velocity Ratio - V.R is defined as the ratio of distance moved by effort to the distance moved by load. No units

Mathematically

$$V.R = \frac{\text{distance moved by Effort}}{\text{distance moved by load.}}$$

Note! V.R is dimensionless.

Efficiency (%) - is defined as the ratio of the useful work done by the machine to the actual work input into the machine.

$$\text{Efficiency (\%)} = \frac{\text{Work output}}{\text{Work input.}}$$

Relationship b/w Mechanical Advantage, Velocity Ratio and Efficiency of Machines.

Speed of train, $V = 80 \text{ km/h}$

$$= \frac{80 \times 1000}{60 \times 60} = 22.22 \text{ m/s}$$

Since the speed of the train is steady, it implies that the pull of the engine equals to the frictional resistance.

$\therefore$ Pull of engine, $F = 28 \times 10^3 \text{ N}$

(i) Power P , developed by the engine is given as

$$P = F \times V$$

$$= 28 \times 10^3 \times 22.22$$

$$= 622 \text{ kW}$$

$$(ii) \text{ Speed } V, \text{ is } \frac{20 \text{ km/h}}{60 \times 60} = \frac{20 \times 1000}{60 \times 60} = 5.56 \text{ m/s}$$

$$\text{acceleration, } a = 1.2 \text{ m/s}^2$$

$$\therefore \text{ Power} = FV$$

$$= Ma \times V \quad (\text{recall } F = Ma)$$

$$= 3000 \times 10^3 \times 1.2 \times 5.56$$

$$= 20 \text{ MW}$$

Torque and Workdone by Torque:

Torque is defined as the product of force and the $\perp$ distance of its line of action from the given Pt or axis. Newton's 2nd law of motion when applied to a rotating bodies state that the torque is directly proportional to the rate of change of angular momentum, Mathematically,

$$\text{Torque } (\tau) = \frac{dL}{dt}$$

Since I is constant

$$\therefore \tau = I \times \frac{d\omega}{dt}$$

but angular acceleration, $a = \frac{d\omega}{dt}$

$$\Rightarrow \tau = I \alpha \quad \text{where } I \text{ is the moment of inertia.}$$

Power: Can be defined as the rate of doing work, and its unit is Watt (W), larger units are in kilowatt (kW) and Megawatt (MW) e.g. etc.

$$\text{Power} = \frac{\text{Work done}}{\text{Time taken}}$$

$$= \frac{\text{Force} \times \text{distance moved by force}}{\text{time taken}}$$

$$= \frac{\text{force} \times \text{distance moved by force}}{\text{time taken}}$$

$$\therefore \text{Power} = \text{force} \times \text{velocity}$$

$$P = F \times V$$

$$\text{Note: } 1 \text{ W} = 1 \text{ J/s} = 1 \text{ N m/s}$$

Worked Example:

1. A force of 1200 N acts on a body moving it through a distance of 5m in the direction of the force in 8 seconds. Determine the Power used.

Solution:

$$\text{Force, } F = 1200 \text{ N}$$

$$\text{Distance moved, } s = 5 \text{ m}$$

$$\text{Time taken, } t = 8 \text{ s}$$

$$\text{Power} = \text{force} \times \text{velocity}$$

$$= \frac{\text{force} \times \text{distance moved (s)}}{\text{time taken}}$$

$$= \frac{1200 \times 5}{8}$$

$$\therefore P = 750 \text{ W}$$

2. A train of mass 3000 tonnes travels along a level track against frictional resistances of 28 kN. cal.

(i) The Power developed by the engine when the train is travelling at a speed of 80 km/h

(ii) The Power developed at the instant that speed is 20 km/h and the acceleration is 1.2 m/s².

Solution:

$$\text{Frictional resistance, } F_r = 28 \times 10^3 \text{ N}$$

$$\text{Mass of train, } M = 3000 \text{ tonnes}$$

$$= 3000 \times 10^3 \text{ kg}$$

Let the height of the block above the ground level be h m.

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$$\therefore P.E = mgh$$

$$\Rightarrow h = \frac{P.E}{mg} = \frac{6.4 \times 10^3}{200 \times 10} = \underline{\underline{3.2 \text{ m}}}$$

Kinetic Energy: Can be defined as the energy possessed by virtue of its velocity. The K.E of a body of mass, m moving on a straight path with velocity v is given by:

$$K.E = \frac{1}{2}mv^2$$

Worked Example:

1. A body of mass 80 kg moves with a constant velocity and covers a distance of 20 m in 5 seconds . Determine the K.E of the body during this motion.

Solution:

Mass of body, $m = 80 \text{ kg}$

Distance covered, $s = 20 \text{ m}$

Time taken, $t = 5 \text{ s}$

Velocity of the body v is given by: $\frac{\text{distance covered (s)}}{\text{time taken}}$

$$\Rightarrow v = \frac{20}{5} = 4 \text{ m/s}^2$$

$$K.E \text{ of the body} = \frac{1}{2}mv^2$$

$$= \frac{1}{2} \times 80 \times (4)^2$$

$$\therefore K.E = \underline{\underline{640 \text{ J}}}$$

2. The K.E possessed by a body of mass 40 kg moving with a velocity v is 500 J . What is the value of the velocity v ?

Solution:
Mass of body, $m = 40 \text{ kg}$

K.E of the body = 500 J

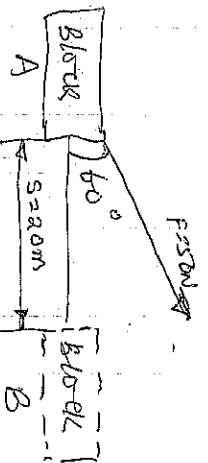
But $K.E = \frac{1}{2}mv^2$

$$mv^2 = 2 K.E$$

$$v^2 = \frac{2 K.E}{m} = \frac{2 \times 500}{40}$$

$$\therefore v = \sqrt{\frac{2 \times 500}{40}} = \underline{\underline{5 \text{ m/s}}}$$

(6)

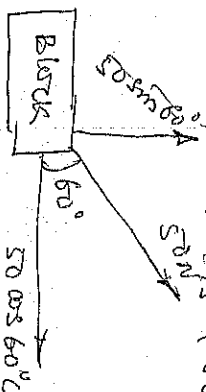
Force applied, $F = 250 \text{ N}$ Distance moved, $s = 20 \text{ m}$ Angle the force makes with horizontal, $\theta = 60^\circ$ Work done $W = F \times s \cos \theta$

$$= 250 \times 20 \cos 60^\circ$$

$$= 5000 \text{ J}$$

When the force is applied to pull the block horizontally, the work done is 1,000 J. But when the force is applied at an angle of 60° to the horizontal, the work done is 500 J.

This implies that less effort is required to pull the block through the same distance when the force applied is inclined to the horizontal. This however can be attributed to the fact that the inclined force has two components as shown below:



Horizontal Components of the force $= 250 \cos 60^\circ$

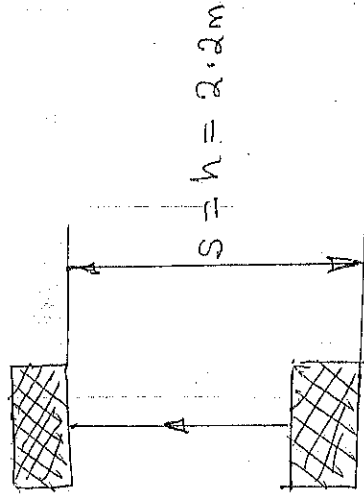
Vertical Components of the force $= 250 \sin 60^\circ$

It can be observed that, the vertical components $= 213.3 \text{ N}$

43.3 N is responsible for lifting the block while the horizontal components of the force 25 N is responsible for pulling the block.

When ever an inclined force is applied to move an object horizontally, we have a combination of lifting and pulling effects on the block ~~which~~ ^{which} makes work done to be less than when the block is pulled by a horizontal force alone.

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Mass of block = 80 kg

Vertical height, $h = 2.2 \text{ m}$

$$g = 9.81 \text{ m/s}^2$$

In this case, the weight of the block is considered the force

$$\therefore W_{\text{block}} = Mg$$

$$= 80 \times 9.81 = \underline{748.8 \text{ N}}$$

Work done on the block W is given by:

$$= W_{\text{block}} \times h$$

$$= 748.8 \times 2.2$$

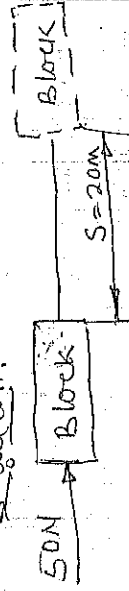
$$= \underline{1.73 \text{ KJ}}$$

Note! the work done on the block is equal to the change in potential energy of the block.

- 4) A force of 50 N is applied to pull a block through a distance of 20 m. Determine the work done under the following conditions:
- (i) when the block is pulled horizontally.
 - (ii) when the force is applied at an angle of 60° to the horizontal.

Hence, compare the answer of (i) and (ii) and explain the disparity.

Solution:



Force applied, $F = 50 \text{ N}$

Distance moved, $S = 20 \text{ m}$

$\therefore$ the work done W is pulling the block is given by:

$$W = F \times S = 50 \times 20 = \underline{1,000 \text{ J}}$$

(If the acceleration is not given, then use the following: (i) centripetal acceleration (ii) Deflection angle (iii) Deflection angle

through a distance s , by the application of the force F is given by' (3)

$$W = F \times d \times \cos \theta$$

Little consideration shows that if F and s are parallel then $\theta = 0$ and $\cos 0 = 1$

$$\therefore W = F \times s$$

Worked Example:

A. Cal. the workdone when a casting is moved 20m along a workshop floor by a force of 3.5kN

Solution:

$$\text{Force applied} = 3.5 \text{ kN} = 3.5 \times 10^3 \text{ N}$$

Distance moved by the casting, $s = 20 \text{ m}$

$$\text{workdone} = F \times \text{distance moved} (s)$$

$$= 3.5 \times 10^3 \times 20 \\ = 70,000 \text{ J}$$

(2) A force of 300N is applied to a body, if the workdone is 6kJ, determine the distance through which the body is moved in the direction of the force.

Solution:

$$\text{force applied } F = 300 \text{ N}$$

$$\text{workdone} = 6 \text{ kJ} = 6 \times 10^3 \text{ J}$$

$$\text{distance } s = ?$$

$$\Rightarrow W = F \times s$$

$$\Rightarrow s = \frac{W}{F} = \frac{6 \times 10^3}{300} = 20 \text{ m}$$

(3) Cal. the workdone in raising 80kg block through a vertical height of 2.2m (Take $g = 9.81 \text{ m/s}^2$)

$K_{rot} = \frac{I}{\sqrt{m}}$ where I is the moment of inertia ⁽³⁰⁾ about the x-axis and m is the mass of the object.

Moment of Inertia: is commonly used in mechanical design calculations. There are basically two types of moment of inertia, they are:

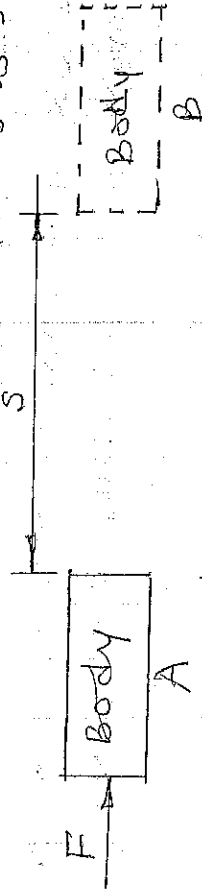
- (i) Area moment of inertia or second moment of area
- (ii) Mass moment of inertia.

Work, Energy and Power (Chapter 4)

Work: is said to be done when a force is applied to a body and causes it to move in the direction of the force. i.e. Workdone = force $\times$ distance moved in the direction of the force. and its units is the Joule (J).

Note: 1 Joule is equal to one newton meter of work.
 $\Rightarrow 1 \text{ J} = 1 \text{ N m}$

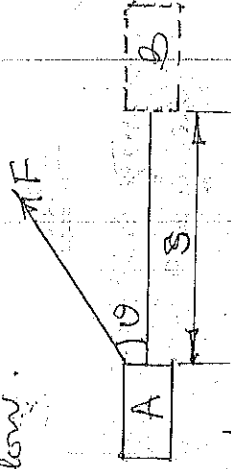
Suppose a constant force F is applied to a body and causes it to move from pt A to pt B through a distance S as shown below



Then the workdone on the body can be expressed as;

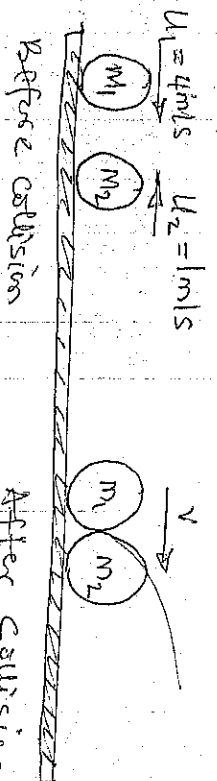
$$\text{Workdone} = \text{Force}(F) \times \text{distance moved}(S) = FS$$

Suppose a force F , is applied to a body at an angle θ , such that the body moves through a distance, S , as shown below.



The work done, W in moving the body from pt A to pt

(28)



By applying the law of conservation of momentum considering the two direction of momentum and total momentum b4 collision = total momentum after

$$\Rightarrow M_1 u_1 + M_2 u_2 = V (M_1 + M_2)$$

collision.

$$\Rightarrow (3 \times 1.5) - (2 \times 2) = V (3 + 2)$$

$$5V = 0.5$$

$$\therefore V = \underline{0.1 \text{ m/s}} \text{ after}$$

Angular Momentum is a vector quantity that represents the product of an object's rotational inertia and rotational velocity about a given axis. It is a function of the object's mass moment of inertia and the angular speed.

Angular momentum is defined as the product of the mass moment of inertia, I , and the angular velocity ω , of a body.

Mathematically, Angular momentum = $I \times \omega$,

$$\text{But } I = mr^2 \text{ and } \omega = \frac{v}{r}$$

$$\therefore L = mr^2 \times \frac{v}{r}$$

$$\Rightarrow \boxed{L = rmv}$$

where L = angular momentum

L is considered as the product of the radius of rotation, r , and the linear momentum, mv .

where V is the tangential (linear) speed at radius

Collision of Bodies:

(26)

There are two types of collision from the subject point of view viz (i) elastic collision (No loss of kinetic energy)

(ii) Inelastic collision

Elastic collision:

$$M_1 u_1 + M_2 u_2 = M_1 v_1 + M_2 v_2 \quad (\text{Momentum is conserved})$$

$$\frac{1}{2} M_1 u_1^2 + \frac{1}{2} M_2 u_2^2 = \frac{1}{2} M_1 v_1^2 + \frac{1}{2} M_2 v_2^2 \quad (\text{Kinetic energy is conserved}).$$

Law of Conservation of Momentum:-

State that in any collision b/w 2 bodies, the total momentum in any direction is unchanged, provided no external force acts in that direction.

It also states that if two or more bodies collide in a closed system, the total momentum b/w collision is equal to the total momentum after collision.

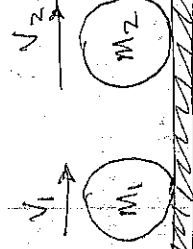
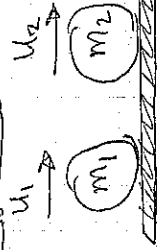
Note: A system is said to be closed or isolated if no external force act on it.

Worked Example:

1. A body of mass M_1 moving with a velocity u_1 collides with another body of mass M_2 moving with velocity u_2 in the same direction as the first body. After collision, they continue in the same direction with velocities v_1 and v_2 respectively.

Applying the law of conservation of momentum, deduce an expression for the system.

Solution:



Before Collision

After Collision

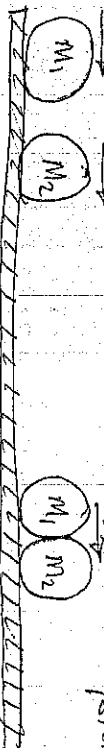
A little consideration will show that there is a closed system since no external force act on it. $\therefore$, applying the law of conservation of momentum gives

Total momentum b/w collision = Total momentum after collision.

$$\Rightarrow M_1 u_1 + M_2 u_2 = M_1 v_1 + M_2 v_2$$

2. A body of mass M_1 moving with a velocity u_1 collides with another body of mass M_2 moving with velocity u_2 in the same direction as the first body. After collision, the two

bodies stick together and move with a common velocity V in the original direction. Applying the law of conservation of momentum, deduce an expression for the system.



Before collision

After collision

* Little consideration will show that, after collision, the two bodies move with common velocity, V , applying the law of conservation of momentum yields:

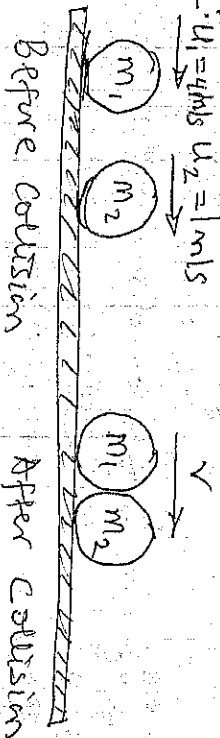
Total momentum before collision = Total momentum after collision

$$\Rightarrow M_1 u_1 + M_2 u_2 = V(M_1 + M_2)$$

where V is the common velocity with which M_1 and M_2 move with after collision.

③ Two bodies, masses 30kg and 20kg travelling at 4m/s and 1m/s respectively in the same direction, collide and continue together, find their common speed after the collision.

Solution: $u_1 = 4\text{m/s}$ $u_2 = 1\text{m/s}$



Before collision

After collision

Applying the law of conservation of momentum yields:

Total momentum before collision = Total momentum after collision

$$\Rightarrow M_1 u_1 + M_2 u_2 = V(M_1 + M_2)$$

collision

$$\Rightarrow (30 \times 4) + (20 \times 1) = V(30 + 20)$$

$$50V = 140$$

$$V = 2.8\text{m/s}$$

④ Two trucks of masses 3kg and 2kg move towards each other at speeds of 1.5m/s and 2m/s. After collision they move together. Find their common velocity.

THE FEDERAL POLYTECHNIC, MUBI
DEPARTMENT OF MECHANICAL ENGINEERING
2ND AND 4TH SEMESTER EXAMINATIONS 2021/2022 ACADEMIC SESSION
COURSE CODE/TITLE: MEC 124, MECHANICAL ENGINEERING SCIENCE II
PROGRAMME: NDI, (MECHANICAL ENGINEERING & AGRIC. ENG. TECHNOLOGY)

TIME ALLOWED:

3 HOURS

INSTRUCTION:

ANSWER ANY FIVE (5) QUESTIONS

ALL QUESTION CARRY EQUAL MARKS

QUESTION ONE

- (a) Define Dynamics and state the two distinct parts of dynamics. (3 mark)
(b) Differentiate between centrifugal and centripetal force. (4 marks)
(c) A 600mm diameter pulley lifts a load at 0.5m/s initially, and over a period of 10 Seconds, the speed is uniformly accelerated to 2m/s. Determine:
(i) The acceleration of the load
(ii) The angular acceleration of the pulley
(iii) The distance moved by the load
(iv) The number of turns made by the pulley (13 marks)

QUESTION TWO

- (a) Define Angular acceleration and derive its relationship with linear acceleration stating the unit of each symbol used. (8 marks)
(b) An electric train moves from rest with a uniform acceleration of 1.5m/s^2 for the first 10Sec and continues accelerating at 0.5m/s^2 for a further 20 seconds. It continues at constant speed for 90 seconds and finally takes 30 seconds to decelerate uniformly to rest:
(i) Draw a graph of velocity against time for the journey
(ii) From your graph, deduce the total distance travelled. (iii) What is the average velocity of the train for the whole journey? (12 marks)

QUESTION THREE

- (a) Define, give the mathematical expressions and state the unit of the following:
(i) Displacement (ii) Velocity (iii) Acceleration (9 marks)
(b) A car test track has a curve of radius 200mm banked at an angle of 17° to the horizontal.
(i) Calculate the maximum speed possible of a car around the bent, if no slide trust occurred between the tyres and the road.
(ii) If the coefficient of friction between the tyres and the track is 0.68. Calculate the maximum speed possible around the bent, assuming skidding occurred. (11 marks)

QUESTION FOUR

- (a) Define, give the mathematical expressions and state the unit of the following:
 (i) Work. (ii) Power (6 marks)
- (b) State the principle of conservation of energy (2 marks)
- (c) A vehicle of mass 1600kg increases its speed uniformly from 36km/h to 72km/h by the action of a resultant force of 2.4 kN. Determine the increase in the kinetic energy of the vehicle during the acceleration period. Show that this increase in kinetic energy is equal to the work done by the accelerating force. (2 marks)

QUESTION FIVE

- (a) Enumerate the classes of 16 lever system and give two (2) examples of each. (4½ marks)
- (b) With reference to simple machines, define and write clearly the mathematical expression for the following:
 (i) Mechanical advantage (7½ Marks)
 (ii) Velocity ratio
 (iii) Efficiency
- (c) A screw Jack has a threaded pitch of 8mm. An effort of 120N applied at the radius of 500mm is just sufficient to move a load of 35kN. Calculate;
 (i) The Mechanical Advantage of the Screw Jack
 (ii) The velocity ratio of the screw Jack
 (iii) The efficiency of the screw Jack (8 marks)

QUESTION SIX

- (c) Define the term Simple Harmonic motion. (2 marks)
- (d) State Newton's first, second and third laws of motion. (9 marks)
- (e) A particle moves with simple harmonic motion in a straight line with amplitude 0.15m and period 18 seconds. Find:
 (i) The maximum speed.
 (ii) The maximum acceleration of the particle
 (iii) Write down the values of the constant P and Q in the equation.
 $x/m = P \sin(Q(t/s))$ which describes its motion. (9 marks)

QUESTION SEVEN

- (a) Define momentum and Radius of Gyration and state their units. (5 marks)
- (b) In a tabular form, differentiate between mass and weight of a body. (6 marks)
- (c) A winding drum of radius 0.75 m is required to raise a lift of mass 1100 kg at a steady velocity of 1.3 m/s. Calculate the power required to drive the drum, neglecting the mass of the cable. Take $g=9.81 \text{ m/s}^2$ (9 marks)

